

SWE 101 – Semester Project Guide Part-2

Submission Deadline: Monday, March 9, 2026

Format: PDF and editable (.docx) files together

1. Team Information

- Elisa-02052548
 - Katerina-02052534
 - Isadora-02052528
 - Blerta-02052510
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- Proposed Scrum Role Rotation Plan (who starts as Scrum Master?):
 - Sprint 1: Elisa – Scrum Master
 - Sprint 2: Katerina – Scrum Master
 - Sprint 3: Isadora – Scrum Master
 - Sprint 4: Blerta – Scrum Master

2. Project Title

.SafeNest – Smart Home Security System

3. Problem Statement (Max 200 words)

Many homeowners lack a reliable and affordable way to monitor their homes in real time. Traditional security systems are often expensive, difficult to access remotely, and do not provide instant alerts. As a result, users may not be aware of security threats until it is too late. This creates a need for a system that provides real-time monitoring, immediate notifications, and remote access to security cameras in a simple and efficient way.

4. Proposed Solution (Max 300 words)

SafeNest is a smart home security system that allows users to monitor their homes through connected cameras and a mobile application. The system provides live video streaming, motion detection alerts, and secure storage of recordings. Users can access the system remotely and manage multiple cameras from one interface. The system is designed for homeowners and small property owners who need an affordable and easy-to-use security solution.

The system will be developed as a mobile and web-based application using a backend server for processing and data storage.

The system will NOT include advanced AI facial recognition or large-scale enterprise security management features, keeping the scope suitable for a semester project.

5. Stakeholders

List at least 4 stakeholders.

For each:

- Stakeholder name/type
- What do they care about?
- What risk do they create?

Example:

Stakeholder	Interest	Risk
Homeowner	Easy monitoring of home	May stop using if UI is complex
Technician	...System maintenance	System downtime issues
Admin	System control	System vulnerabilities
Guest user(optional)	Limited access	Privacy concerns

6. High-Level Requirements

6.1 Functional Requirements (5–8 items)

FR1: The system shall allow users to log in securely

FR2: The system shall allow users to view live camera feeds

FR3: The system shall send motion detection alerts to users

FR4: The system shall allow users to add and remove cameras

FR5: The system shall allow users to view recorded footage

FR6: The system shall support multiple cameras per user

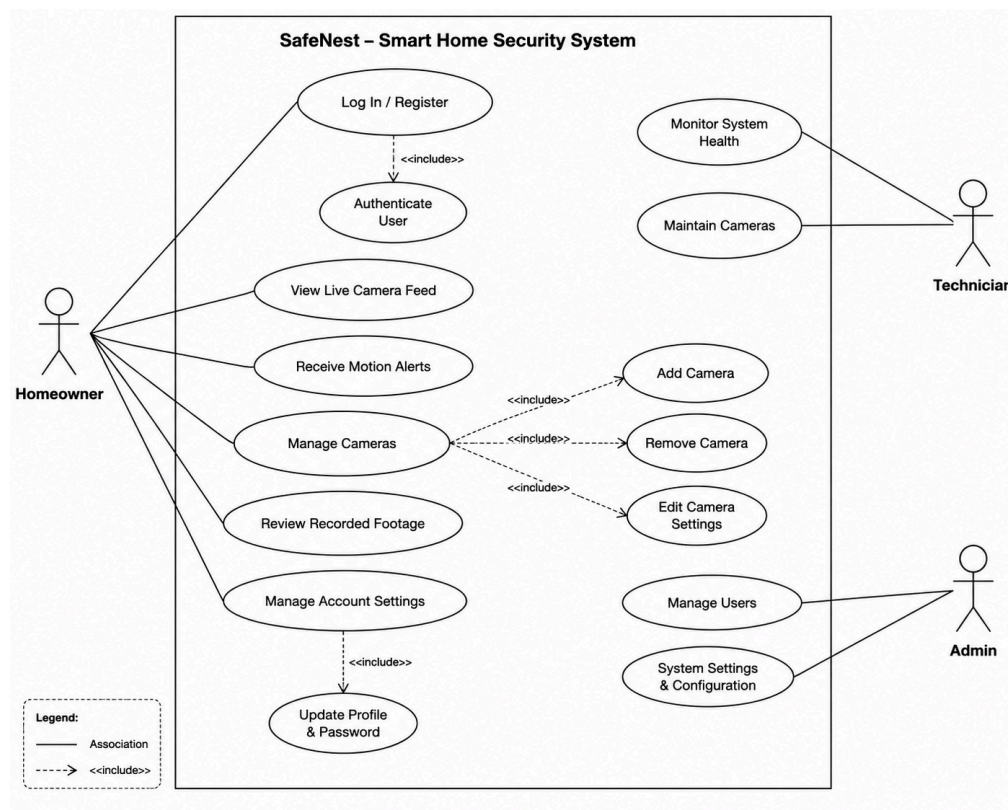
6.2 Non-Functional Requirements (At least 3)

System response time shall be under 2 seconds for normal actions

The system shall ensure secure data encryption for user data

The system shall be available 24/7 with minimal downtime

7. Initial System Model (Very Simple)



8. Technology Stack (Tentative)

Programming Language: Java

Framework: Spring Boot

Database: MySQL

Version Control: Git & GitHub

Hosting: Local / Cloud (optional)

Justification:

These technologies are widely used, beginner-friendly, and suitable for building scalable backend systems.

9. Risks (At least 3)

Risk	Likelihood	Mitigation
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Unequal team contribution	Medium	Divide tasks clearly
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Time management issues	High	Follow sprint plan strictly
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Technical difficulties	Medium	Use simple technologies
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Scope too Medium Limit features early
large

10. Development Plan (High Level)

Sprint 1: Requirements + system design

Sprint 2: UI mockups + diagrams

Sprint 3: Final report + testing + improvements

11. Ethical & Social Considerations

The system handles personal security data such as camera feeds. Misuse of this data could lead to privacy violations. To prevent this, the system will use secure authentication and encrypted data storage. Access to video feeds will be restricted to authorized users only.